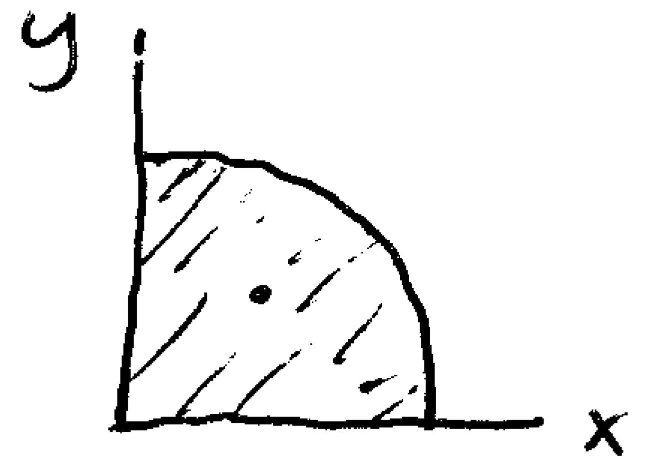
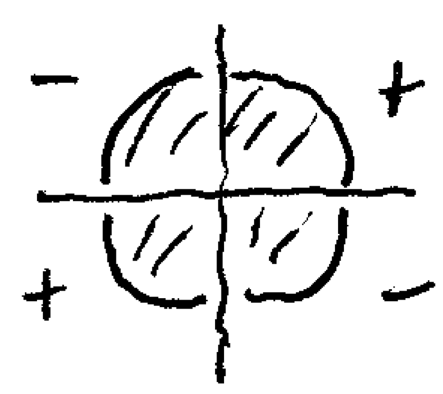
Working I_{xy} (POI) problems for composite areas using Official Crib Sheet formulas.

For these shapes, Hibbeler gives I_{xy} in quadrant 1 only.





$$\boxed{I_{xy} = +\frac{1}{8}R^4}$$

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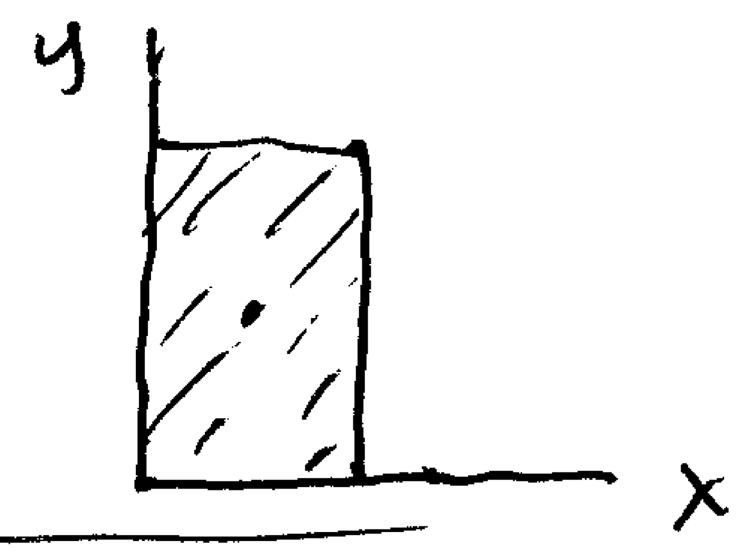
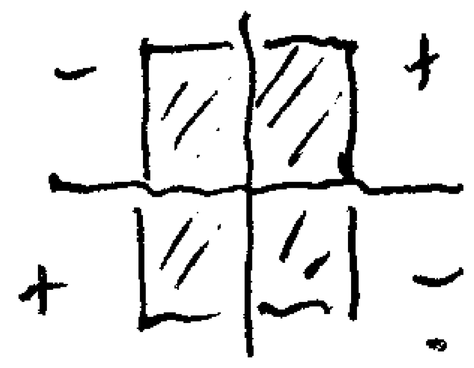
* x-y are <u>not</u> centroidal axes as we used.





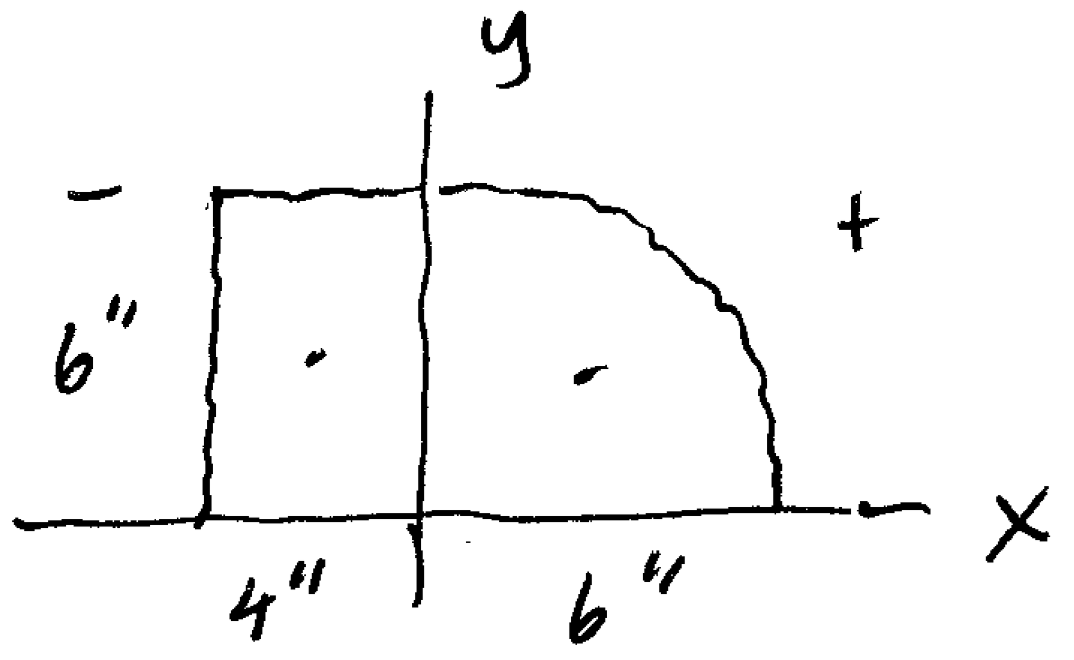
$$\boxed{I_{xy} = +\frac{1}{4}b^2h^2}$$

If have a composite area made up of these shapes, where these shapes <u>touch</u> <u>both</u> <u>the</u> <u>x and y axes</u>, then can use these formulas directly. But must use the correct sign!

Do not have to use the full parallel axis theorem (PAT) since they <u>touch</u> both x + y and you have an I_{xy} formula to use directly.

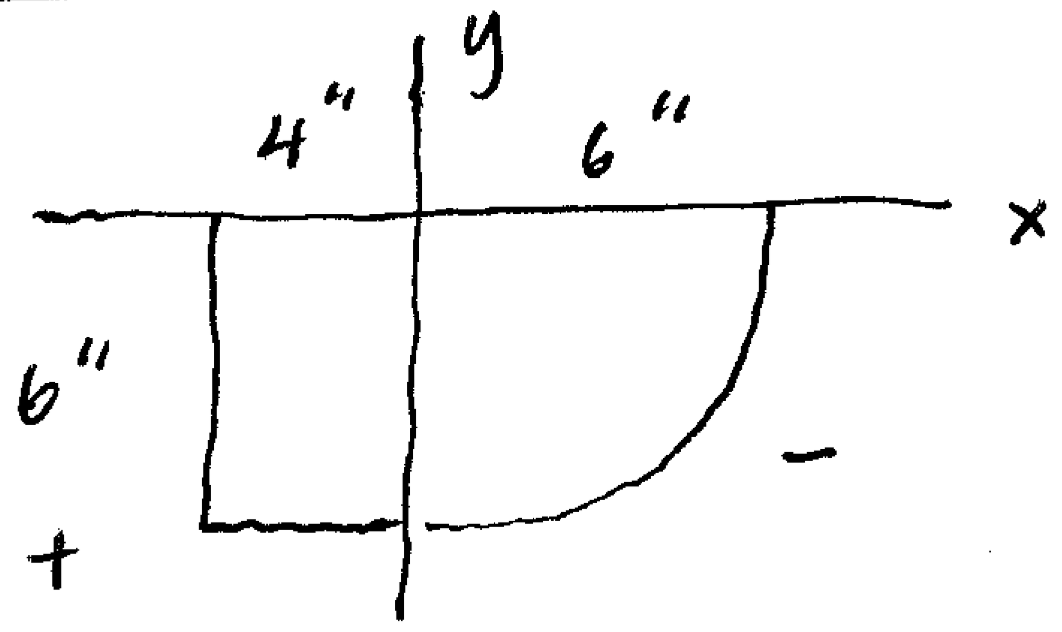
Examples

①



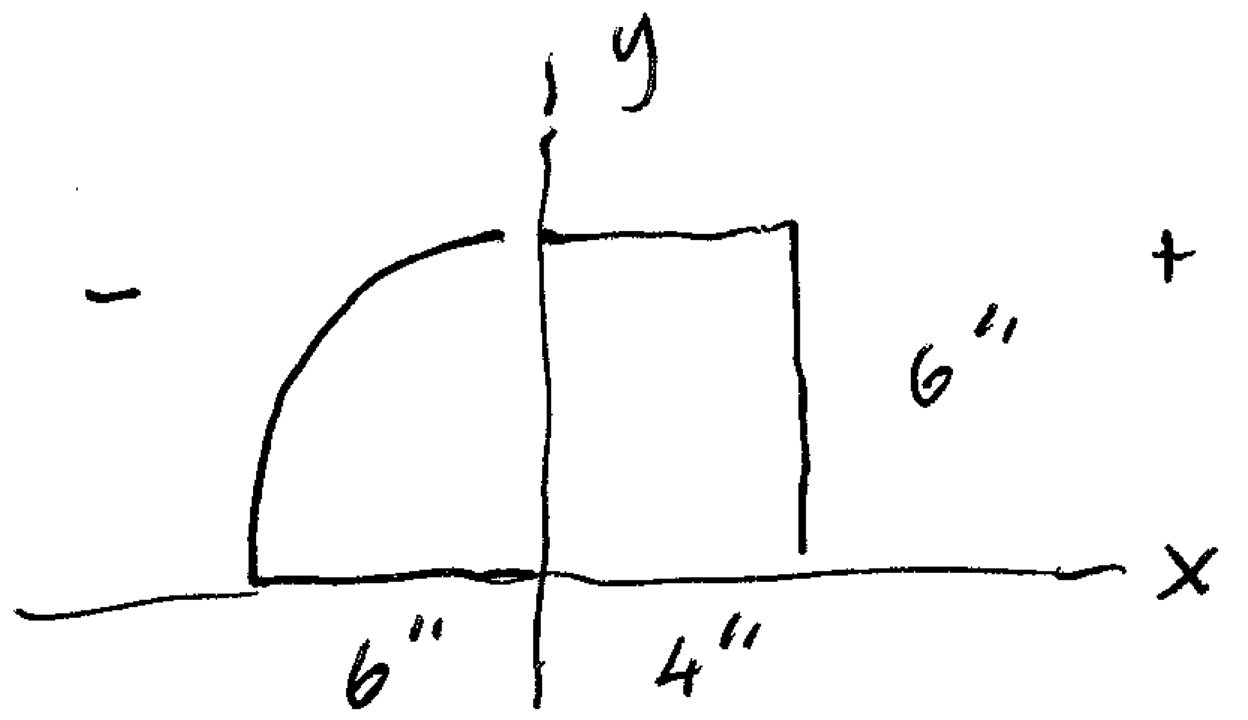
$$I_{xy} = +\frac{1}{8}(6)^4 - \frac{1}{4}(4)^2(6)^2$$

②



$$I_{xy} = -\frac{1}{8}(6)^4 + \frac{1}{4}(4)^2(6)^2$$

③



$$I_{xy} = -\frac{1}{8}(6)^4 + \frac{1}{4}(4)^2(6)^2$$

etc.